New factor-base leads

Small chart grafts, cofactor fibres and witness diversity

2026-10-19 · finite native structural screen

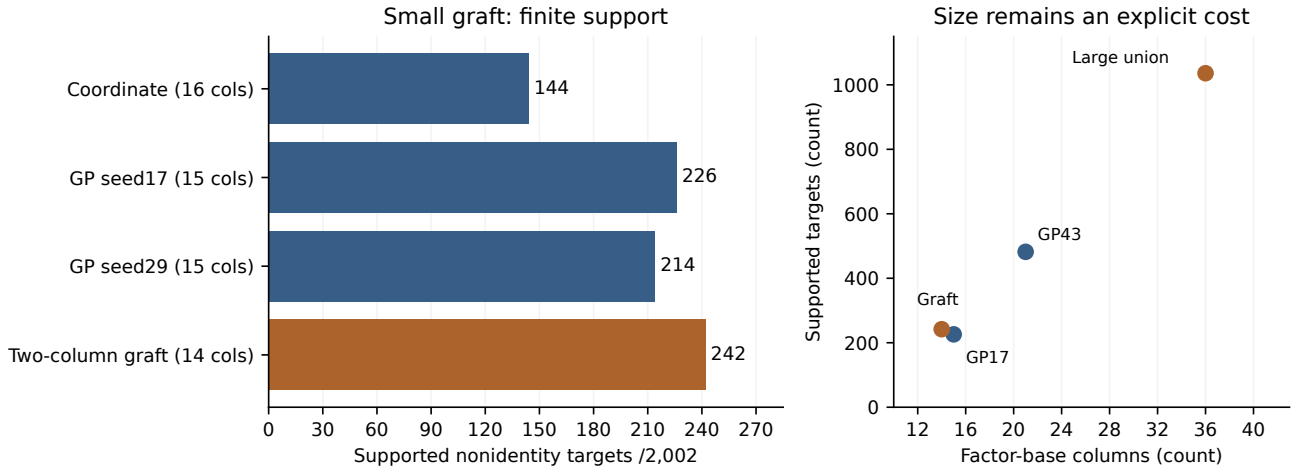
A concrete smaller-base candidate

For `icv1-f2m13-t181-515ee569`, a 14-column union supports **242** of 2,002 nonidentity subgroup targets. Earlier GP seed 17 supports **226** with 15 columns. The union grafts a two-column auxiliary chart onto a 12-column base. This example was selected after inspection; it has no fresh holdout test.

Case	Columns	Support	Collision rank
Coordinate B029	16	144	0
GP17 B031	15	226	3
GP29 B035	15	214	4
Graft B027+B033	14	242	3

Rank means differences between alternative witnesses for the same target, modulo r, after removing zero prime-projection columns. It is not training rank or rank per unit of online work. The graft raises finite support but does not raise this rank relative to GP17.

icv1-f2m13-t181-515ee569 · exact structural census



42 distinct nonempty bases; 313 unions; all pair memberships checked natively.
Post-hoc graft example; chart branches, successful F4 work, full pricing and regularity are unmeasured.

Observed: the graft's two separate supports cover 200 targets; 42 more need cross-chart witnesses.
Proposed: test the four signed auxiliary points by point subtraction and exact main-base membership, retaining the main-main solver. This branch is not implemented or priced in this study.

No claim: successful F4 work, certified regularity, runtime speedup, statistical significance, full pipeline cost or wide-curve applicability.

What the new screen checked

Protocol/source commit `4bce792529a2ed81103a91809f65853e9ccde213`; standalone lock pinned in `6811c867f8b86aa3880d48aa8ddaa5f37efed89c` before execution. The 60 inventories came unchanged from the completed 2026-10-18 study: one empty failed inventory retained, 59 nonempty records, 42 exact distinct signed point sets. Every within-curve two-base union was included, for 313 unions and 355 total cases. No 256-signed-point cap was reached. Repeated union sets are correlated cases, not independent replicates.

Exact ICV1 model	r	Bases	Unions	Pair checks
`icv1-f2m7-tm13-ac10a42c`	71	17	136	4,278
`icv1-f2m9-tm5-4a3ea183`	37	3	3	465
`icv1-f2m13-t181-515ee569`	2,003	19	171	43,365
`icv1-f2m31-tm90707-c95f16f5`	1,439,393	3	3	14,878

All 62,986 unique unordered point pairs (repetition included) were added as actual curve points. Direct multiplication of each sum by r decided subgroup membership. Identity sums were counted separately. The cofactor-image test agreed for every pair. Every completed singleton reproduced the prior independent Rust census. All targets and witness pairs are retained.

The arithmetic adapts the published independent audit's polynomial-basis field/group implementation. It is not a new independent arithmetic implementation or independent-host certificate. Exact generators, EC1/full UID, subgroup, field, FB1 identities and derivations remain in copied inputs. No scalar logarithms were searched for or recovered.

Preregistered selections and their limits

Per curve: most new cross targets; fewest columns with any new cross target; highest useful witness-difference rank, with deterministic ties. All 355 compact cases and these selections appear in `summary.json`.

The maximum-cross degree 13 union `B031+B041` uses 36 columns and supports 1,036 targets; 654 already occur in the union of singleton supports, 382 are new cross targets. Its rank is 35. The largest useful-rank selection is `B034+B041`, 38 columns, support 758, rank 36. Size and rank are different objectives. At 21 or fewer columns, no union exceeds the 21-column GP43 singleton's support 482. The smaller graft on page1 is explicitly post-hoc.

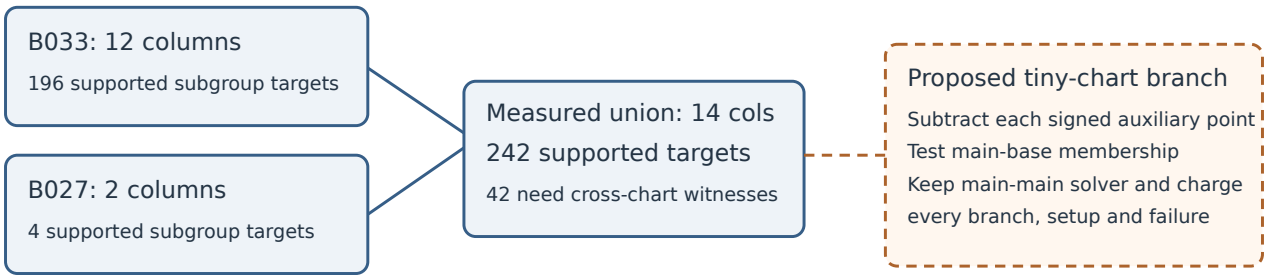
Execution and accounting

Card `CARD-chart-complements-20261019-1` executed one exact native command, with 70 bindings and 900-second timeout, exit 0. The director assessed `executed_as_specified`, relay verbatim. Tool elapsed time is an operational receipt, not a benchmark. The shared managed host earned no timing isolation; only deterministic structural counts are interpreted.

An initial tool-process refusal occurred before native execution. Post-run CLI relay checking hit an output-as-input lint defect; unchanged canonical verification functions checked the receipt instead. Exact failure and fallback records are preserved. No native result was retried.

Three implementation leads

icv1-f2m13-t181-515ee569 — observed inventory union and unmeasured next branch



Solid boxes: exact finite census. Dashed box: not implemented or priced; no regularity or speed claim.

1. A tiny-chart graft with an explicit membership fast path

B033 (Frobenius-squared dimension 5 seed 17) has 12 columns/support 196; B027 (trace-zero/Frobenius dimension 4 seed 29) has 2 columns/support 4. Their union has 14 columns/support 242, 42 mixed-only targets. Freeze a new matched-target protocol before implementing the proposed branch. Count point subtraction, membership, setup, failures, new log columns and all other arms. A general two-chart solver has up to four ordered branch cases, or three with justified symmetry handling; this is not free extra support.

2. Search for useful cofactor-fibre collisions

On `icv1-f2m31-tm90707-c95f16f5`, coordinate B000 has 68 signed points and 68 distinct images under multiplication by r. Its 34 identity pairs yield zero nonidentity subgroup pairs. Random B001 similarly has 48 points, 48 images, 24 identity pairs and zero support. Their 58-column union supports only2 targets. Coordinate plus B002 supports 4 with 62 columns. Every degree 31 case has useful homogeneous rank 0.

Proposed filter: seek low-degree chart parameters that make compatible fibres contain more than inverse pairs. Require actual subgroup support before online solving; charge the construction and scalar checks as setup. Projection by the cofactor may destroy simple membership. No degree 83/130 confidence gate or transfer has been performed.

3. Admit witnesses for rank and cost, not yield alone

The same-target collision matrix and active witness-pair graph are retained. Graph connectivity alone does not guarantee rank; rank alone does not give cheap successful decomposition. Proposed next step: compare first-witness and rank-aware admission on fresh targets, including the cost of enumerating alternative roots. No online selector ran here, and old draft cards remain unrun, zero-funded and unleased.

Mathematical status

This is a finite checked census, not a new general theorem or polynomial identity. No Lean proof, IDC1 certificate, regularity certificate or formal verification of the arithmetic is claimed. Direct scalar checks decide support; a cofactor identity is not assumed as a substitute. See the named obstruction `problems/factor-base-shape-to-successful-work.md`.

Validation, reproducibility and prior art

Standalone native checks: 5 tests passed; strict all-target Clippy, formatting and release build passed. 9 CI-scope tests passed; naming: 0 problems. Draft card validates with two intentional TODO warnings.
Required root library command: ONE complete serial release-library command: 4,400 passed, 0 failed, 118 existing ignores, 0 filtered. Same previously bound binary; no fresh root compilation claimed.
Existing root inputs: All 1,632 existing source 819 native input hashes match before and after. Unit binary SHA256 8ce288f3bfcc1529dc63f91d71bdffa852226582fd3bef110b26e789e8261e1f.
Local dedicated CI replay: Both exact dedicated CI Bash steps pass locally; full census output byte-identical. Archive member hashes and deterministic extracted-tree repacking pass. GitHub results pending publication.
Coordination: Conductor absent 127; pinned workflow 404; neither is a pass.

The five tests cover field inverses, repeated scalar addition, inverses and associativity branches, modular-rank dependencies and exhaustive small-curve cofactor checks. Strict all-target Clippy and release build pass. All input and source hashes, raw member hashes and extracted-tree deterministic repacking are checked. Current source/records, failures and operational limitations are in `VALIDATION.md` and `SETUP_OBSERVATIONS.md`.

Rebuild the standalone research executable from `screen/Cargo.toml`, then invoke `factor-base-chart-screen screen STUDY_ROOT NEW_OUTPUT_JSON` directly. Compare the result with `runs/census.json` after extracting `runs.tar.xz`. Check the archive sidecar, `runs/SHA256SUMS` and `SOURCE_SHA256SUMS` first. The dedicated CI workflow repeats this; local validation is not a GitHub pass. The installed unified `factor-base-sweep` executable and its resume bindings were not replaced. This study's verifier is not a new public launcher.

Relevant primary literature

Gorla–Massierer, Index calculus in the trace zero variety (2014/318). Their extension-degree 3/5 setting supplies context, not validation of the present degree 31 fixture or prime-degree wide challenges.

Faugère–Huot–Joux–Renault–Vitse, Symmetrized summation polynomials (EUROCRYPT 2014). Torsion symmetries can compact decomposition systems in the studied regimes. A fibre histogram is not that implemented solver. Abstract retrieved through primary archive search; direct PDF open returned 403 this session.

Euler–Petit, New results on quasi-subfield polynomials (2019/2021). Structured-root-set families and their restrictions suggest a separate membership direction; the abstract does not establish a resulting ECDLP speedup. Not tested here. No global novelty claim is made for any lead.

What remains open

Successful solver work for chart unions, fresh-target incremental rank, complete symmetric native-operation pricing, matched fastest external tools, certified regularity and large-field scaling remain unmeasured. The new follow-up card stays draft/unrun/zero-funded/unleased, with no runnable frozen protocol or checkpoint. No S3 operation, paid host, historical Python rerun, new campaign through old cards or modification to frozen evidence occurred.

The canonical IC cost scoreboard, progress timeline, curve/leaderboard data and prior frozen plots remain unchanged: no admitted performance incumbent or end-to-end claim changed. No timing benchmark entry is appropriate for these structural counts. Complete artifact links and requirements status appear in the study README.